

Summary

Identification Strategy

1. Data and Treatment Classification
2. Exposure Construction
3. Descriptive Statistics
4. Credibility Checks
5. Core Confirmatory Results
6. Robustness
7. Exploratory Analyses
8. Multiple Testing and Inferential Hierarchy
9. Limitations
10. Conclusion
11. Machine Learning & Neural Network Extension
12. Wage Pass-Through

State Minimum Wage Increases and Low-Wage Employment

A Difference-in-Differences Study with Power, Heterogeneity, and Robustness Analysis

September 20, 2026

Summary

This report estimates the employment effect of the 2021 round of state minimum-wage increases on food service and retail employment, and asks whether that effect scales with pre-policy wage exposure. It follows a pre-specified analysis plan (Section 2 of the project proposal): a headline average-effect result (Model A), a primary scientific hypothesis about exposure heterogeneity (Model C's β_4 , explicitly pre-registered as likely underpowered), and a full confirmatory battery – event study, placebo test, COVID-sensitivity spec, wild cluster bootstrap, and model diagnostics – run before either headline number is taken at face value.

The honest headline: the credibility checks found a real problem, not a clean result. Pre-COVID-era trends between treated and control states were not parallel for food service employment, confirmed by two independent methods (the event study and a simple two-sample t-test), and the entire baseline effect is highly sensitive to how differently states recovered from COVID – a covariate for that alone reduces Model A's estimated effect by roughly 90%. This is reported as a direct limit on how much weight the average-effect estimate can carry, exactly as the proposal's own design anticipated it might have to be (Section 6.5).

The exposure gradient (β_4) has its own, separate credibility problem. Four independent checks added after the original build – a placebo test, a permutation test, a specification curve, and a dynamic event study (Section 8.1) – all find that high- and low-exposure states already differed systematically before the 2021 policy. β_4 is reported throughout exactly as designed, but it should be read as confounded by this pre-existing association, not as clean evidence either for or against a real exposure gradient.

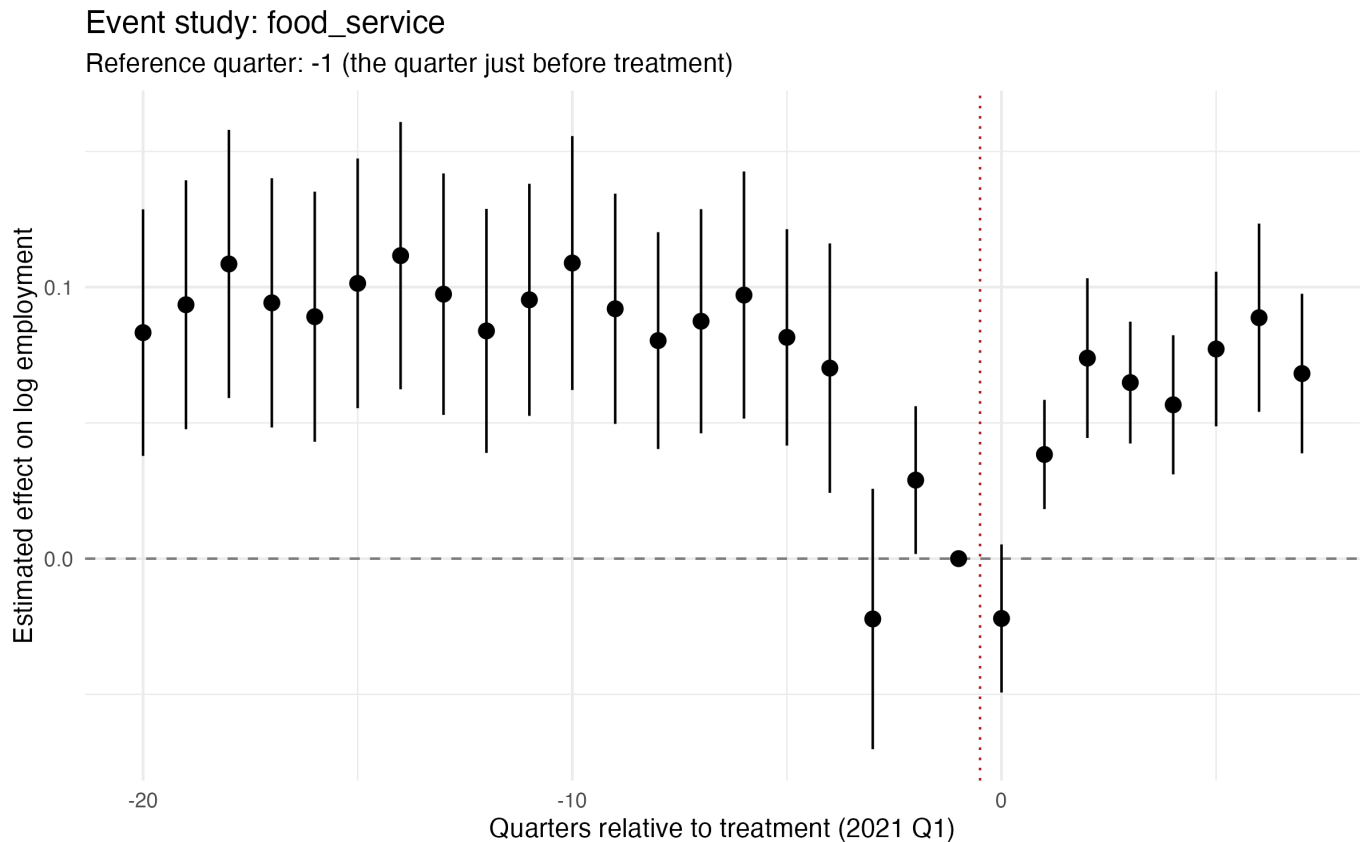
Identification Strategy

Before any regression equation, the causal logic this design relies on, stated in plain language:

- **Treatment.** States with a minimum-wage increase effective in Q1 2021.
- **Comparison.** States with no minimum-wage change at all in 2021.

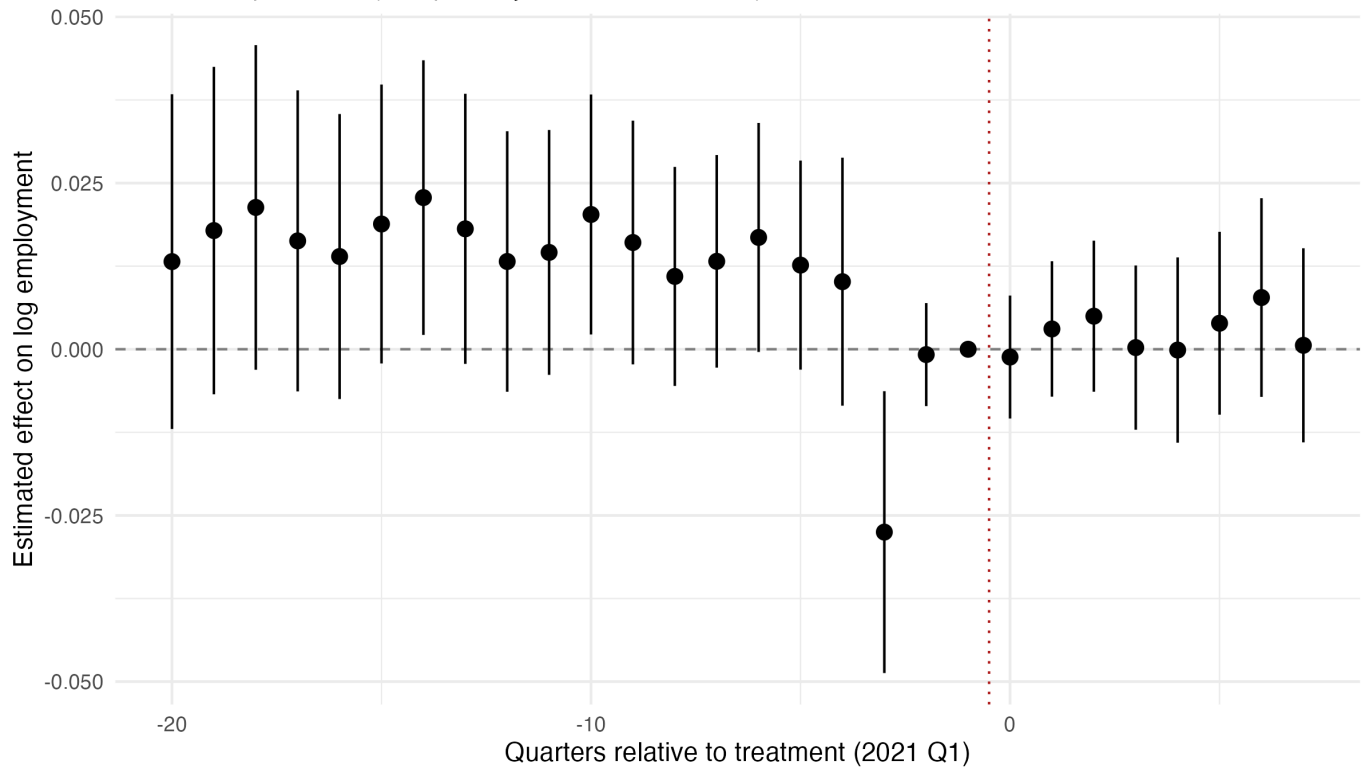
- **Outcome.** Quarterly log employment, food service and retail, tracked separately.
- **Counterfactual.** Absent the minimum-wage increase, treated states would have followed the same employment trajectory as the comparison states – the parallel-trends assumption every DiD design rests on.
- **Main threat.** The 2021 treatment window sits immediately after an extraordinary, unevenly-distributed labor-market shock. If COVID caused states to recover on different trajectories for reasons unrelated to minimum-wage policy, and that difference happens to correlate with treatment status, the counterfactual above fails.
- **Evidence.** Three independent checks, run before any DiD coefficient is reported: an event study of the dynamic pre- and post-treatment effects, a joint statistical test of whether the pre-treatment coefficients are consistent with zero, and a COVID-sensitivity specification that directly tests whether differential recovery explains the baseline result.

Figure 1 – Pre-treatment trends and dynamic treatment effects



Event study: retail

Reference quarter: -1 (the quarter just before treatment)



The formal joint test that every pre-treatment lead equals zero **rejects flat pre-trends for both industries** (food service $F = 4.77$, $p < 0.0001$; retail $F = 3.09$, $p = 0.001$).

Because the pre-treatment coefficients are not jointly consistent with zero, the parallel-trends assumption is not credible for this outcome. Consequently, the baseline DiD coefficient should not be interpreted as a clean causal estimate. Section 4 below tests why directly (a COVID-recovery confound explains a substantial share of it), and Section 9's Limitations section states this plainly rather than qualifying it away. Every result that follows in this report is presented with that finding already on the table, not as a caveat added at the end.

1. Data and Treatment Classification

Source	Role
FRED (public API)	State-quarter employment (food service, retail), GDP, population, 2015-2022
U.S. DOL minimum wage history (Wayback Machine snapshots)	Treatment classification
QCEW Open Data API	Fallback employment series (2 states); wage-ratio validation check
CPS-ORG via IPUMS-CPS	Exposure construction (2019-2020 earnings only)

The design compares **20 treated states** (raised wages effective 1/1/2021) against **25 control states** (source-checked against DOL snapshots from Dec 2020, Feb 2021, and Dec 2021 – confirmed zero minimum-wage movement across all of 2021). A further 5 states raised wages later in 2021 and are excluded from the primary comparison (they aren't a clean control group for a binary treated indicator).

Two corrections surfaced during data acquisition, both source-verified rather than assumed:

- **Michigan's 2021 increase never happened.** The draft treatment table had it going \$9.65 → \$9.87; DOL's own Feb 2021 snapshot still shows \$9.65. Michigan's automatic adjustment is conditioned on the prior year's unemployment rate staying under 8.5%; the COVID-era spike tripped that clause and the scheduled step was skipped.
- **The control group itself was missing initially.** The proposal names the 20 treated and 5 excluded states explicitly but never lists the actual "no 2021 change" control group its own design requires – without it there's no DiD to run. All 25 remaining states

were source-checked and used.

State counts by group

group	n_states
control	25
excluded	5
treated	20

2. Exposure Construction

Exposure is the CPS-ORG-weighted share of a state-industry cell's workers earning within 12.5% of that state's pre-2021 minimum wage, computed **only** from 2019-2020 earnings so the post-period outcome can't contaminate the covariate. The proposal states the band as "10-15%" without picking one number and separately flags band width as a source of construction risk (Section 13); 12.5% (the midpoint) is the primary figure used throughout, with 10% and 15% also computed for sensitivity.

Two validation checks, both passing:

- **Exposure-distribution check:** food service exposure is higher than retail in 49 of 50 states (mean 0.307 vs. 0.196, Wilcoxon $p = 9.93e-10$) – the two industries genuinely differ in exposure before that difference is used in Model C.
- **QCEW correlation check:** QCEW's average wage (hourly proxy) over the state minimum wage, correlated against the CPS-ORG exposure share. Both industries show the expected negative relationship (food service $r = -0.82$, retail $r = -0.71$) – high-exposure states really do have wages sitting closer to the floor.

3. Descriptive Statistics

Pre-period (2019 Q1 -> 2020 Q4) employment trend, by group / industry / increase type

group	industry	increase_type	n	mean_pre_trend	sd_pre_trend
control	food_service	no_change	25	-0.1183	0.0843
control	retail	no_change	25	0.0066	0.0411
treated	food_service	inflation_adj	9	-0.1544	0.0814
treated	food_service	inflation_adj_paused	1	-0.2650	NA
treated	food_service	legislation	10	-0.2286	0.0854
treated	retail	inflation_adj	9	0.0143	0.0161
treated	retail	inflation_adj_paused	1	-0.0143	NA
treated	retail	legislation	10	-0.0207	0.0328

4. Credibility Checks

Per Section 6.5 of the proposal, these are presented and interpreted **before** any DiD coefficient: "the 2021 treatment window follows an extraordinary labor-market shock. This is a central threat to identification... a limit on how much weight the Model A headline number itself can carry if pre-trends look poor."

4.1 Event Study / Parallel Trends

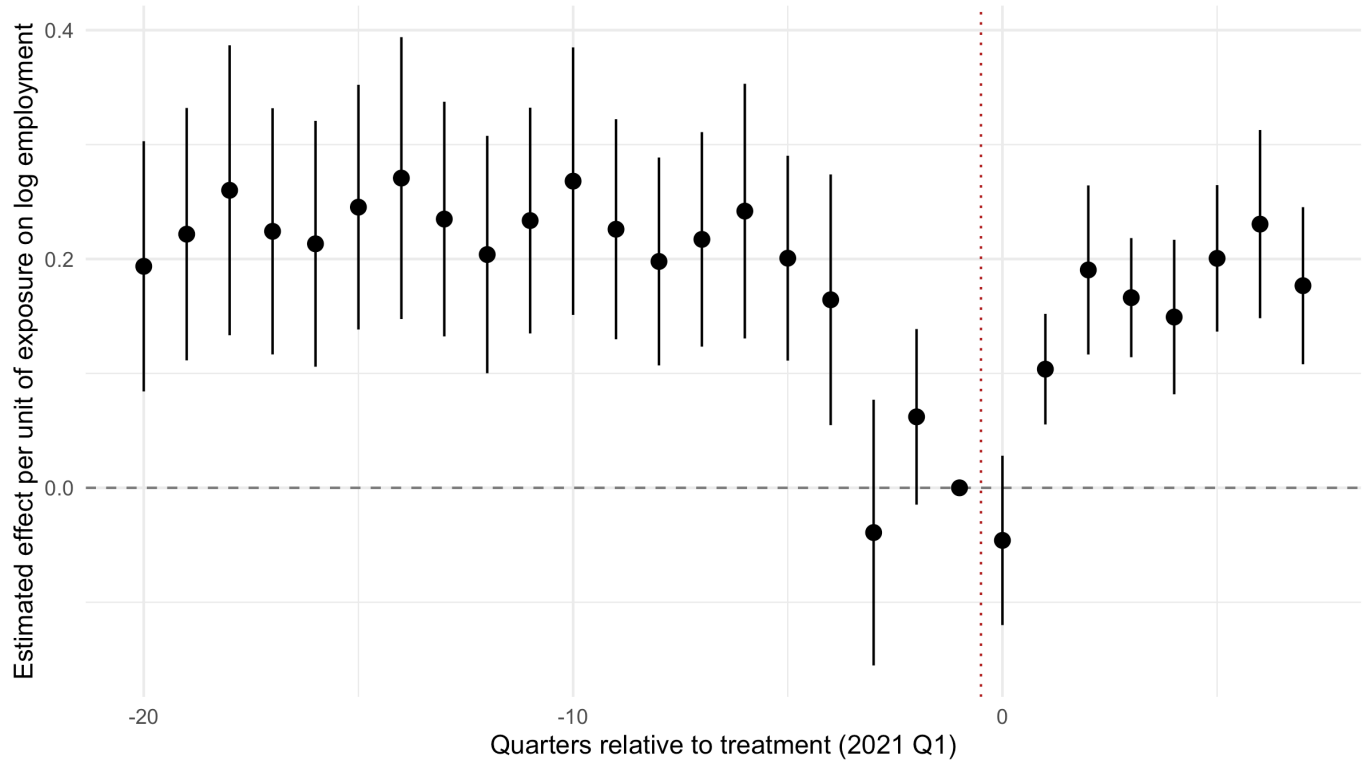
Dynamic treated $\times$ quarter-relative-to-treatment effects, state and quarter fixed effects, state-clustered SEs, 2016 Q1-2022 Q4 (the longest window the growth covariates allow). Shown as Figure 1 in the Identification Strategy section above, ahead of any DiD coefficient; repeated here in a testing context rather than re-embedded.

The pattern isn't a gradual pre-trend drift so much as a roughly stable, non-zero gap between treated and control states across most of the sample, plus one sharp anomalous dip at quarter -3 relative to treatment – **2020 Q2, the COVID employment trough** – most visible in retail.

Post-build addition: the same dynamic check for β_4 . The event study above only ever covered Model A's average effect. Model C's exposure gradient (β_4 , Section 5.2) gets the identical treatment: each relative-quarter dummy interacts with `treated x exposure` (continuous) instead of `treated` (binary), so each coefficient is the dynamic exposure gradient at that quarter rather than a dynamic average effect.

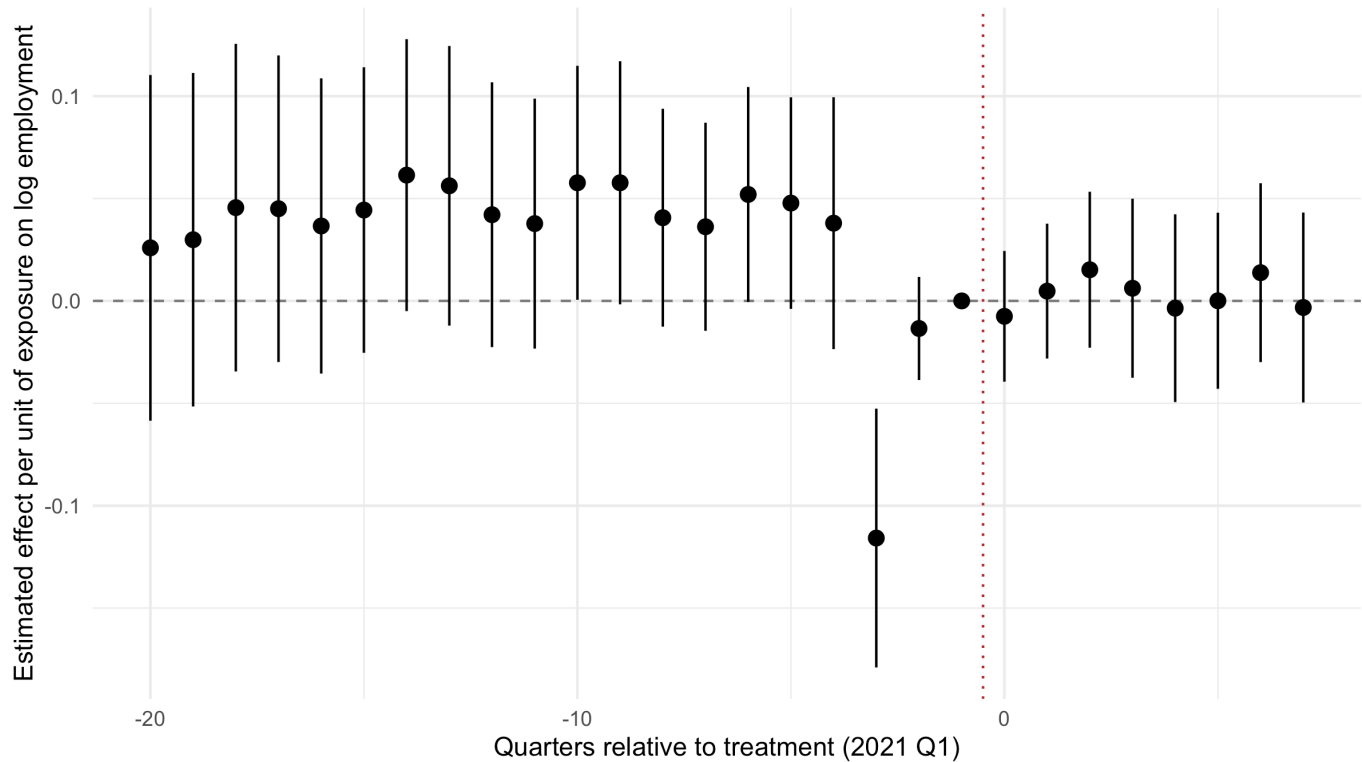
Event study (beta4, exposure gradient): food_service

Reference quarter: -1 (the quarter just before treatment)



Event study (beta4, exposure gradient): retail

Reference quarter: -1 (the quarter just before treatment)



The joint pre-trend test rejects here too, more strongly than Model A's (food service $F = 4.44$, $p = 2.24e-05$; retail $F = 6.76$, $p = 9.18e-08$). The plots show why directly: the exposure-gradient coefficient sits at a roughly constant, clearly positive (food service) or noisy (retail) level across the *entire* pre-period, and doesn't visibly shift after the actual treatment date – high- and low-exposure states were already diverging years before 2021, for reasons this design can't attribute to the real policy. This is the same conclusion the placebo test (Section 4.4) and permutation test (Section 7.3) each reached independently; the event study just makes it visible quarter by quarter rather than at a single false date or through a randomization null. Four separate checks now agree that β_4 should not be read as clean causal evidence.

4.2 Two-Sample t-Test (independent corroboration)

A simpler, STAT-240-level check using a completely different method: compare each state's raw pre-period (2019 Q1 → 2020 Q4) employment trend between treated and control groups directly, no fixed-effects model involved.

Two-sample t-test, pre-period trend: treated vs. control

industry	mean_treated	mean_control	t_statistic	df	p_value	ci_low	ci_high
food_service	-0.1971	-0.1183	3.0259	39.8853	0.0043	0.0262	0.1314
retail	-0.0046	0.0066	1.0485	42.7953	0.3003	-0.0103	0.0328

Food service shows a real, significant difference: treated states' pre-period employment trend was already more negative than control states' *before* treatment started ($p = 0.0043$). This independently confirms the event study's pre-trend violation with unrelated methodology – not a new problem, the same one caught twice.

4.3 COVID-Sensitivity Specification

Adds `covid_severity x post` – each state-industry's log employment change from 2019 Q4 to the 2020 Q2 trough, interacted with the post indicator, since a bare state-constant term would be fully absorbed by the state fixed effect – to Model A and Model C.

Baseline vs. COVID-sensitivity spec

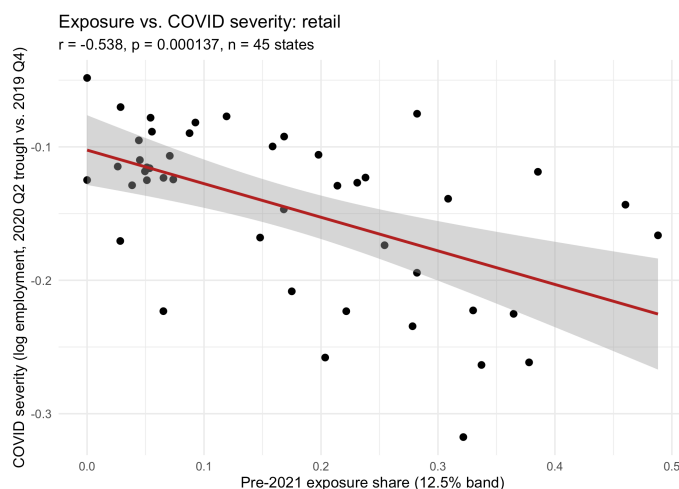
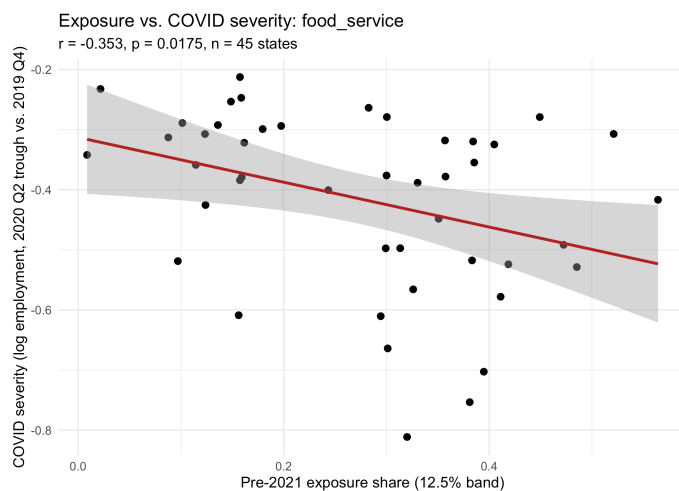
industry	model_a_treated_post_baseline	model_a_treated_post_covid_spec	model_c_beta4_baseline	model_c_beta4_covid_spec
food_service	-0.025	-0.0021	0.1669	0.0626
retail	-0.010	0.0060	-0.0047	0.1018

`covid_severity x post` is highly significant in every specification ($p < 10^{-8}$). Once included, Model A's `treated_post` shrinks toward zero (food service: $-0.025 \rightarrow -0.0021$) and Model C's β_4 shifts substantially, even flipping sign for retail ($-0.0047 \rightarrow 0.1018$). Differential COVID recovery explains a real share of what the baseline specification below picks up.

Post-build addition: does exposure itself predict COVID severity? Section 8.1 finds that high- and low-exposure states already differed on outcome trends before 2021, across four independent checks, without identifying a mechanism. This checks the obvious candidate directly: correlate each state's pre-2021 exposure share with its own COVID severity (same definition as above), restricted to the same treated+control sample every confirmatory model uses.

Correlation: pre-2021 exposure share vs. COVID severity

industry	correlation	p_value	n_states
food_service	-0.3525	0.0175	45
retail	-0.5383	0.0001	45



Both correlations are negative and statistically significant (food service $r = -0.353$, $p = 0.0175$; retail $r = -0.538$, $p = 1.37 \times 10^{-4}$): states with more pre-2021 exposure were hit *harder* by COVID, not just differently. This is a real, plausible candidate mechanism for the Section 8.1 finding – but a candidate mechanism only counts once it’s actually tested, not just proposed.

Does controlling for it resolve the placebo failure? The direct test: add `covid_severity x post` (this section’s own COVID-sensitivity covariate) to the placebo-date Model C specification (Section 4.4) and see whether the spurious placebo β_4 shrinks toward zero.

Placebo β_4 , with and without controlling for `covid_severity x post`

industry	baseline_beta4	baseline_beta4_p	covid_controlled_beta4	covid_controlled_beta4_p	pct_shrinkage
food_service	0.1113	0.0001	0.1020	0.0001	8.3758
retail	0.0640	0.0164	0.0699	0.0144	-9.3188

It largely doesn’t. Food service’s placebo β_4 shrinks by only 8.4% and stays highly significant ($p = 1.26 \times 10^{-4}$); retail’s actually grows slightly rather than shrinking. **The honest conclusion is narrower than the correlation alone would suggest:** pre-2021 exposure does predict COVID severity, but that specific mechanism, controlled for directly, does not explain away the placebo failure. Something about high-exposure states’ pre-2021 employment trends is systematically different from low-exposure states’ for reasons this design doesn’t fully capture – COVID severity is one contributing, correlated factor, not the whole explanation. The Section 8.1 conclusion (β_4 confounded by a pre-existing association) stands on its own regardless of which underlying variable ultimately explains it.

4.4 Placebo Test

A false treatment date (2018-01-01), restricted to a pre-COVID window (2016-2019) so it can’t just re-detect the confound already found above.

Placebo test, Model A (beta3), false treatment date 2018-01-01

industry	placebo_date	placebo_treated_post	placebo_se	placebo_p
food_service	2018-01-01	-0.0055	0.0051	0.2917
retail	2018-01-01	-0.0043	0.0054	0.4268

Clean result for the average effect: small, non-significant estimates for both industries. The design doesn't manufacture a Model A-style effect from nothing when there's no real policy change to detect – the pre-trend problem above is real and specific to the actual treatment window, not a general flaw in the method.

The same check on Model C's exposure interaction (β_4 , Section 5.2, this report's primary hypothesis) tells a different story:

Placebo test, Model C (beta4), false treatment date 2018-01-01

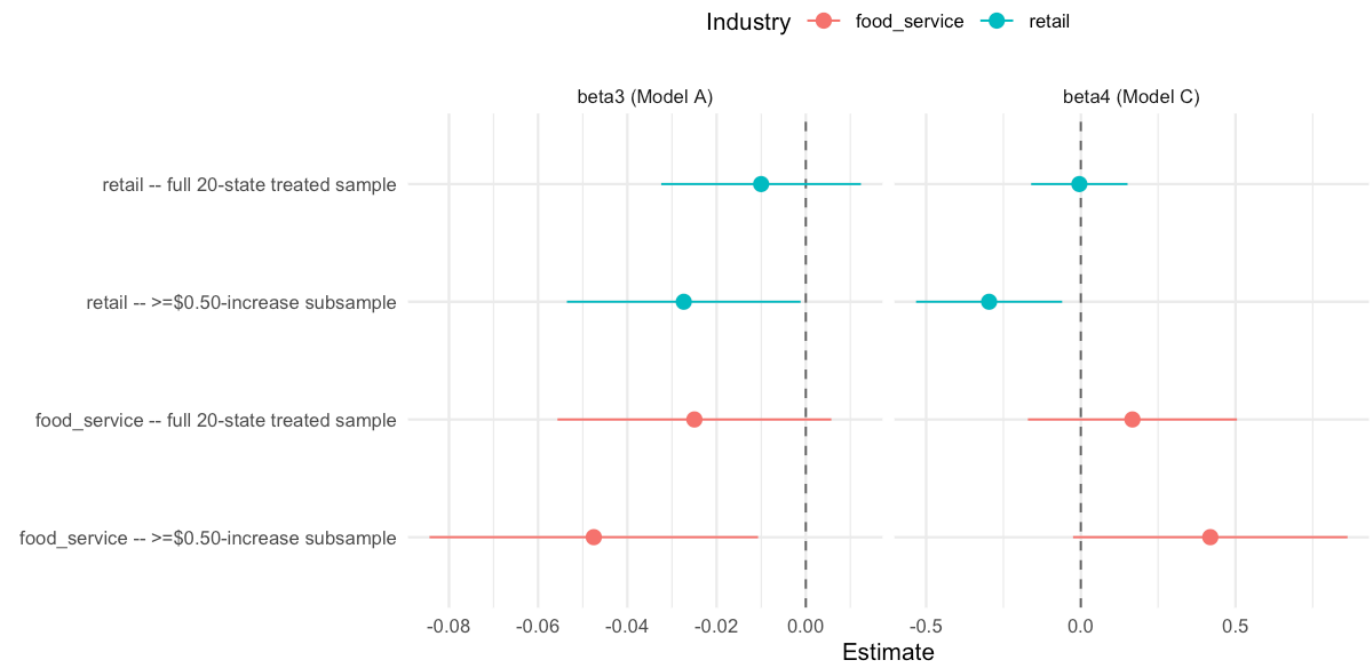
industry	placebo_date	placebo_beta4	placebo_beta4_se	placebo_beta4_p
food_service	2018-01-01	0.1113	0.0257	0.0001
retail	2018-01-01	0.0640	0.0256	0.0164

Both industries come back significant at the fake date (food service $p < 0.001$, retail $p = 0.016$) – the opposite of the clean Model A result above. This means states with higher pre-2021 exposure already had systematically different employment trends in 2016-2019, for reasons that have nothing to do with the real 2021 minimum-wage increases. Since that same exposure measure is what the real Model C relies on to identify β_4 , this is a direct threat to the exposure-gradient result's credibility specifically – not just a general caveat, but evidence that part of what Section 5.2's β_4 estimate is picking up may be a pre-existing structural difference between high- and low-exposure states rather than a treatment effect. Combined with Section 6.3's specification curve, this is the second independent piece of evidence that the exposure-gradient result should not be read as clean causal evidence.

5. Core Confirmatory Results

Given Section 4 above, these are reported with the caveat already attached, not as a clean average effect. All four confirmatory estimates, at a glance, before the specification-by-specification detail:

All confirmatory estimates: beta3 and beta4, both samples, both industries
95% CI. Confirmatory family per Section 8 of the report.



5.1 Model A – Headline Average Effect (β_3)

Two-way fixed-effects DiD: $\log_employment \sim \text{treated_post} + \text{gdp_growth} + \text{pop_growth} \mid \text{state} + \text{quarter}, \text{state-}$
clustered SEs, treated + control states only.

Model A: treated_post coefficient

sample	industry	model_a_treated_post	model_a_se	model_a_p
full 20-state treated sample	food_service	-0.0250	0.0157	0.1179
full 20-state treated sample	retail	-0.0100	0.0114	0.3853
$\geq \$0.50$ -increase subsample	food_service	-0.0475	0.0188	0.0162
$\geq \$0.50$ -increase subsample	retail	-0.0274	0.0134	0.0486

Negative in all 4 specifications, but only significant in the $\geq \$0.50$ -increase subsample (food service $p = 0.016$, retail $p = 0.049$) – the full sample is diluted by treated states whose increases were as small as \$0.00-0.19. Every `fixest` estimate was cross-checked against an independent `lm() + sandwich::vcovCL()` fit; coefficients and clustered SEs matched to 4+ decimal places in all 4 specifications.

5.2 Model C – Exposure Interaction (β_4 , primary scientific hypothesis)

Model C: beta4 (treated_post x exposure)

sample	industry	model_c_beta4_treated_post_x_exposure	model_c_beta4_se	model_c_beta4_p
full 20-state treated sample	food_service	0.1669	0.1724	0.3384
full 20-state treated sample	retail	-0.0047	0.0796	0.9530
$\geq \$0.50$ -increase subsample	food_service	0.4186	0.2264	0.0731
$\geq \$0.50$ -increase subsample	retail	-0.2965	0.1205	0.0191

Noisy and sign-inconsistent across specifications – positive for food service, negative for retail, significant only in the $\geq \$0.50$ subsample. Section 4.4 of the proposal pre-registers this exact possibility: the design may not reliably distinguish “no gradient” from “real gradient” given the sample size. A noisy β_4 is a valid, expected outcome under that pre-registration, not a failed analysis.

5.3 Coefficient Interpretation

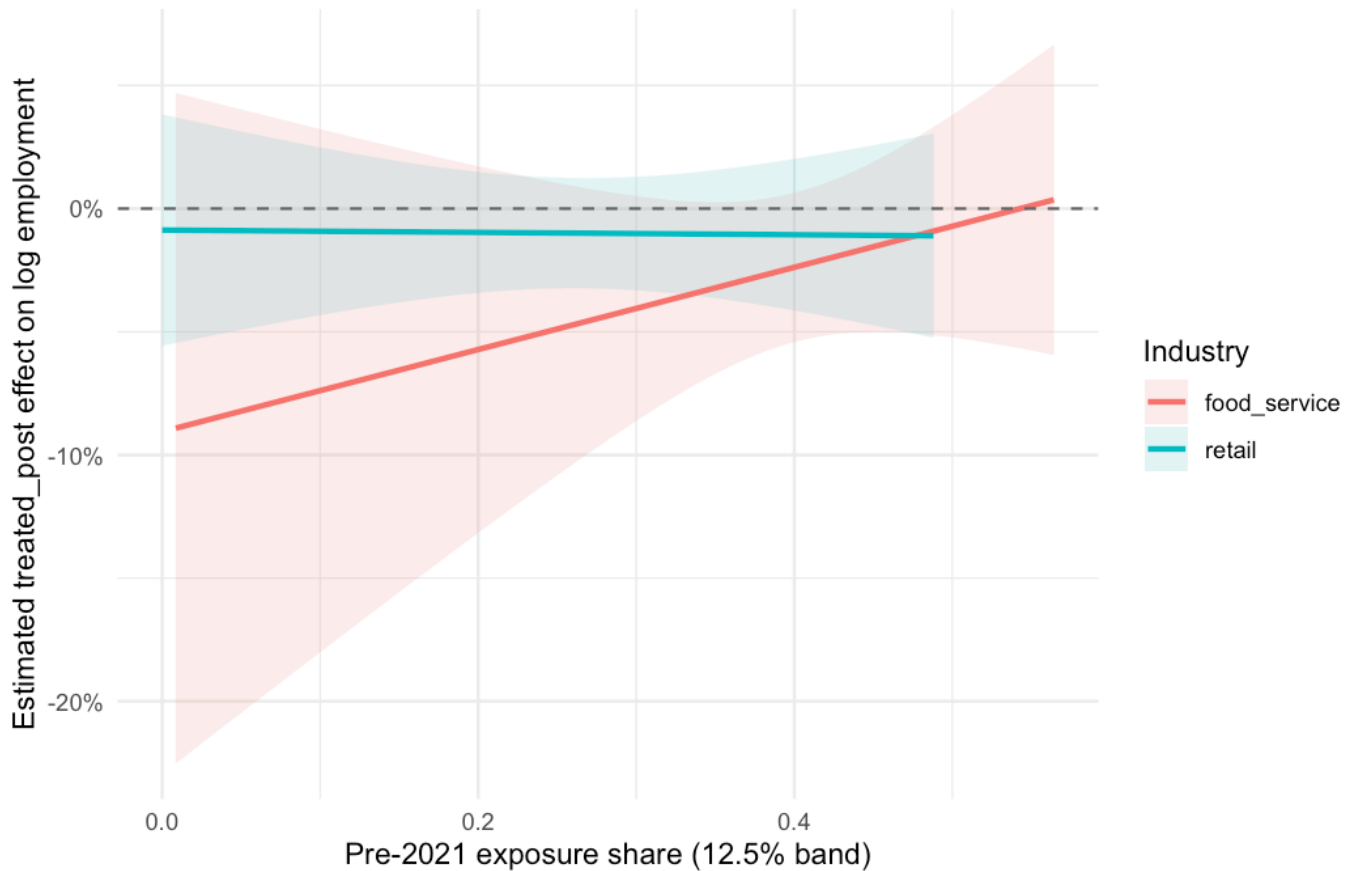
Neither coefficient is self-explanatory from the table alone, so spelled out explicitly:

- **Model A (β_3).** The full-sample food-service estimate is -0.025 – because the outcome is log employment, this corresponds to an approximately 2.5% **lower** employment level for treated states relative to control states after the 2021 increase, holding GDP growth, population growth, and the state/quarter fixed effects constant. It is not a dollar figure, a job count, or a percentage-point change in an employment rate – purely a relative log-scale gap between the two groups.
- **Model C (β_4).** β_4 measures how the *estimated treatment effect itself* changes as a state-industry’s pre-treatment exposure share increases – formally, $\partial(\text{treatment effect})/\partial(\text{exposure}) = \beta_4$. A positive β_4 (food service, this report) means the treatment effect becomes more positive as exposure rises; a negative β_4 (retail) means it becomes more negative. β_4 is not itself an employment effect – it’s the *slope* of the employment effect with respect to exposure, which is why Section 5.4 plots it as a curve rather than a single number.

5.4 Marginal Effect of Treatment by Exposure Level

Estimated treatment effect by pre-2021 exposure level

Model C: $\text{effect}(\text{exposure}) = \beta_3 + \beta_4 \times \text{exposure}$, with 95% CI (delta method)



This is Model C's practical content: the estimated treatment effect ($\beta_3 + \beta_4 \times \text{exposure}$) across the actual range of pre-2021 exposure observed, with a 95% confidence band from the model's clustered variance-covariance matrix (delta method). Food service's line has a visible upward slope – higher-exposure states show a smaller (less-negative) estimated effect – while retail's is nearly flat. The band's width, not just its slope, is the point: it comfortably contains zero across the entire exposure range for both industries, which is the same underpowering conclusion Section 7.5's formal power analysis reaches by simulation instead of visually.

5.5 Exposure Bandwidth Sensitivity

Is β_4 an artifact of the arbitrary 12.5% exposure-band choice? Model C refit at the two alternative bandwidths R/04 already computes (10%, 15%) but never previously used downstream:

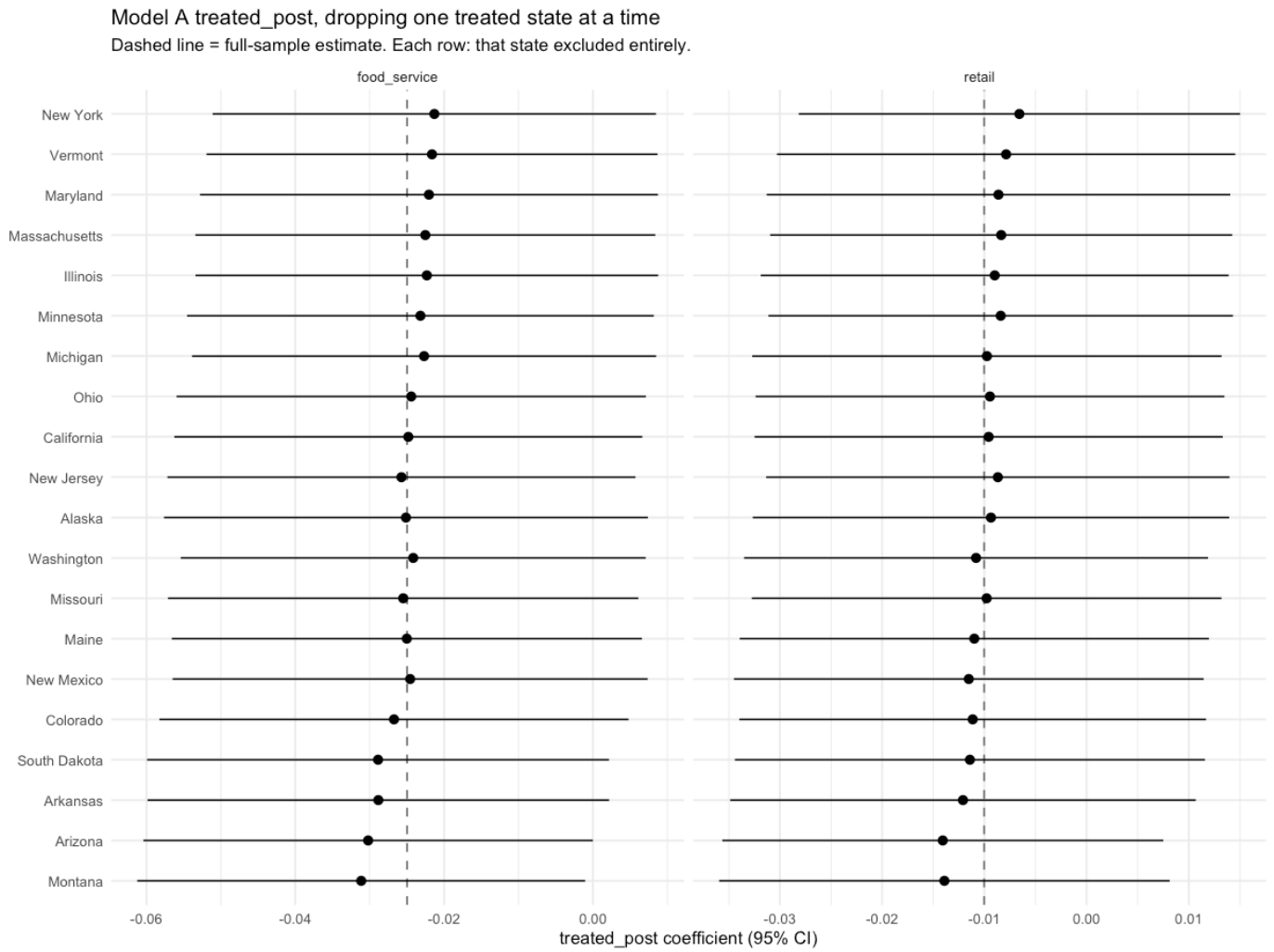
Model C beta4 across exposure bandwidths

bandwidth	industry	beta4	beta4_se	beta4_p
10%	food_service	0.2066	0.1295	0.1177
12.5%	food_service	0.1669	0.1724	0.3384
15%	food_service	0.1086	0.1921	0.5748
10%	retail	-0.0146	0.0898	0.8715
12.5%	retail	-0.0047	0.0796	0.9530
15%	retail	-0.0194	0.0806	0.8107

Food service stays positive and never significant across all three bandwidths; retail stays small and never significant, though its sign flips between the 10% band and the wider two. Nothing here overturns Section 5.2's conclusion – the result is not an artifact of picking 12.5% specifically, but it was never precise enough to distinguish a real gradient from noise at any of the three definitions either.

5.6 Leave-One-State-Out Sensitivity

Is Model A's estimate driven by any single treated state? Section 6.2's model diagnostics already flag Vermont, New York, and Hawaii as high-leverage (2020 Q2-Q3 observations) – this checks what actually happens to the *coefficient*, not just the leverage diagnostic, refitting Model A once per treated state with that state dropped entirely.



Range of treated_post across all 20 single-state exclusions

industry	min_estimate	max_estimate
food_service	-0.0311	-0.0213
retail	-0.0141	-0.0066

The estimate stays negative and moves only modestly regardless of which state is dropped, for both industries – reassuring on its own terms, though it doesn't touch the pre-trend violation or COVID confound already established above.

6. Robustness

6.1 Wild Cluster Bootstrap for β_3 and β_4

`fwildclusterboot`, the standard package for this, was archived by CRAN in 2024 and doesn't build on current R, so the Cameron-Gelbach-Miller procedure (999 Rademacher-weight replications on the restricted-model residuals) is implemented directly, for both coefficients – the proposal's Section 8 only specified β_4 , but the same low-cluster-count concern applies equally to β_3 .

Wild cluster bootstrap, beta4 (full 20-treated-state sample)

industry	beta4	beta4_se_asymptotic	beta4_t_asymptotic	wild_boot_p_value	wild_boot_ci_low	wild_boot_ci_high
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industry	beta4	beta4_se_asymptotic	beta4_t_asymptotic	wild_boot_p_value	wild_boot_ci_low	wild_boot_ci_high
food_service	0.1669	0.1724	0.9678	0.3674	-0.1921	0.5383
retail	-0.0047	0.0796	-0.0593	0.9530	-0.1829	0.1721

Wild cluster bootstrap, beta3 (full 20-treated-state sample)

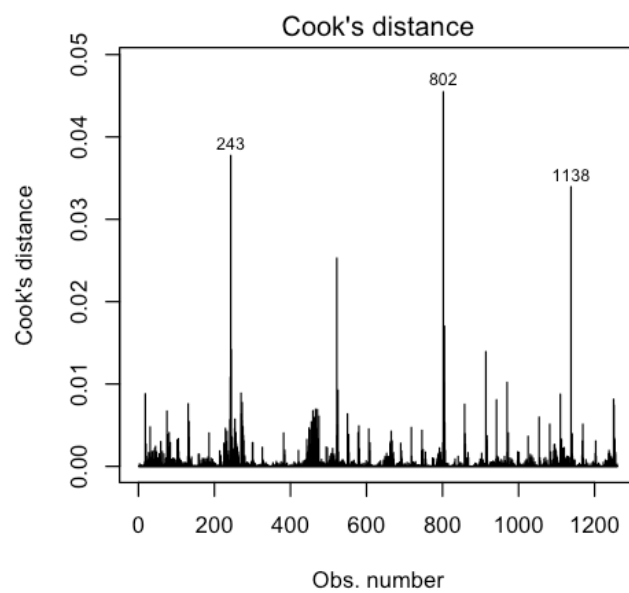
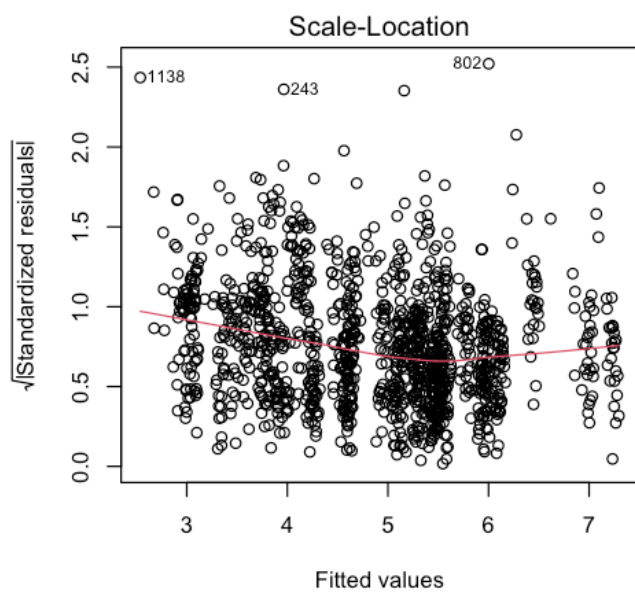
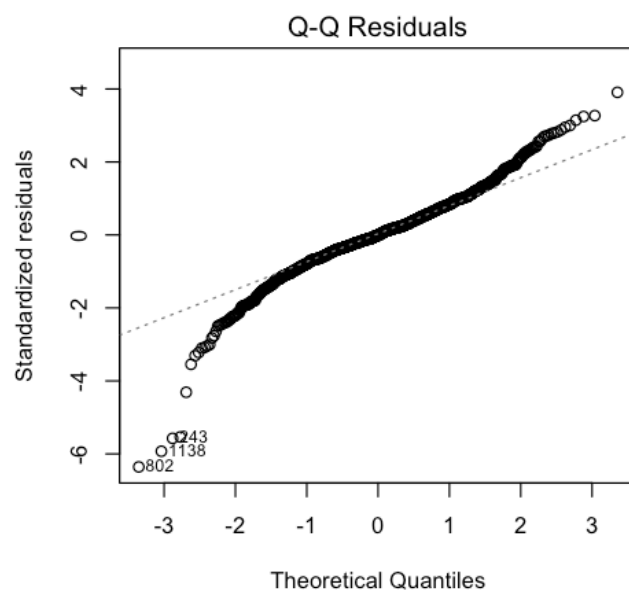
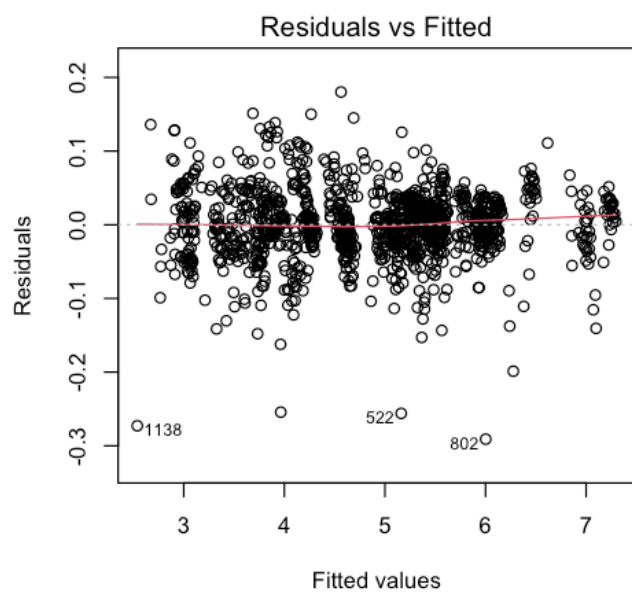
industry	beta3	beta3_se_asymptotic	beta3_t_asymptotic	wild_boot_p_value	wild_boot_ci_low	wild_boot_ci_high
food_service	-0.025	0.0157	-1.5948	0.1071	-0.0560	0.0062
retail	-0.010	0.0114	-0.8770	0.3874	-0.0339	0.0124

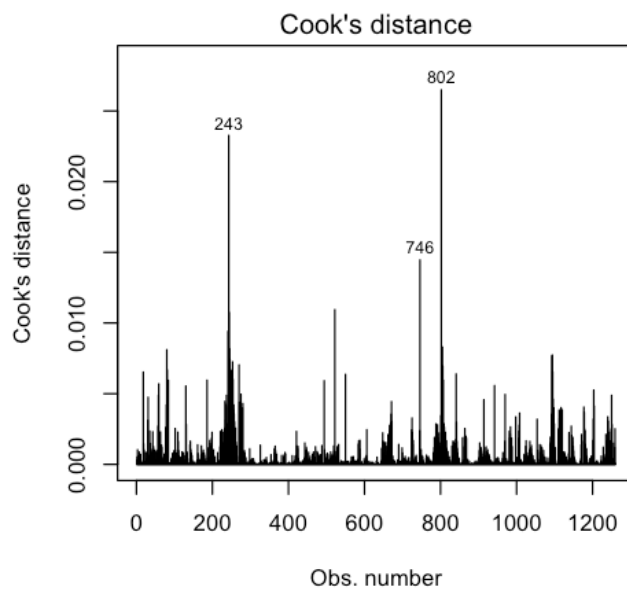
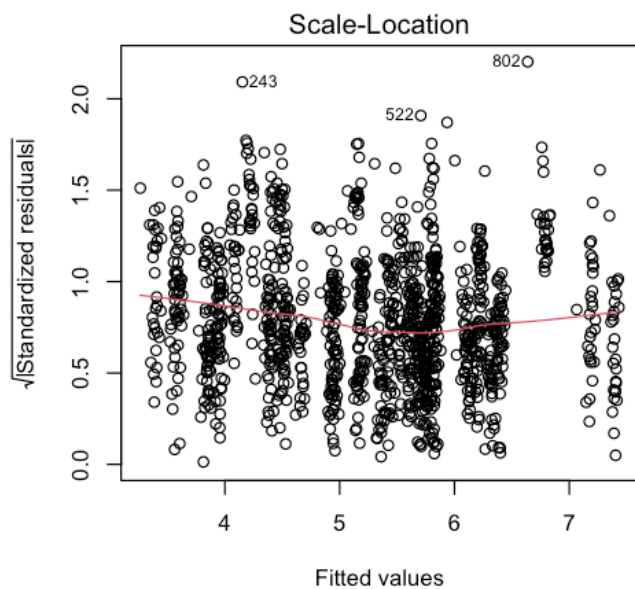
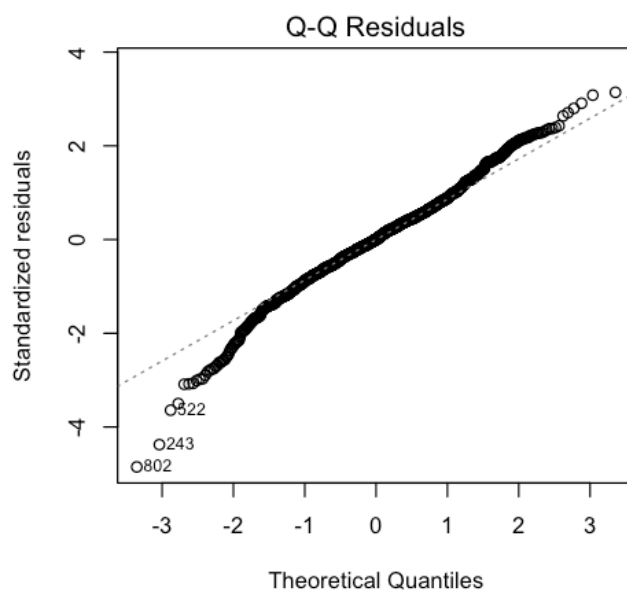
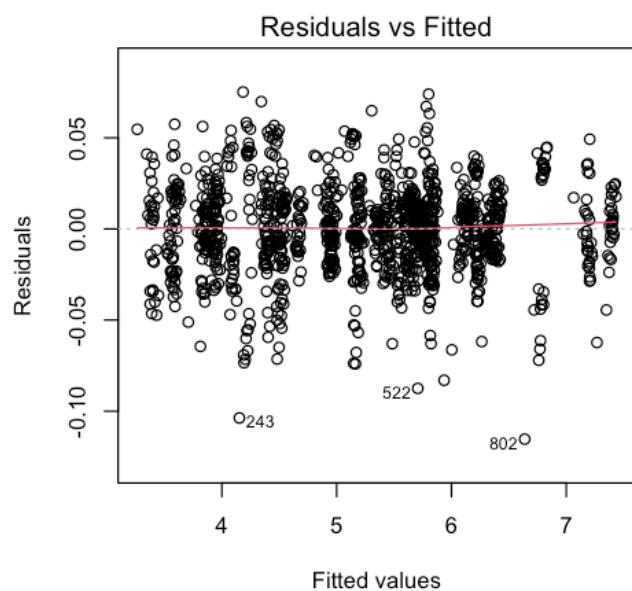
Bootstrap p-values track the asymptotic ones closely for both coefficients (β_4 : food service 0.367 vs. 0.338 asymptotic, retail 0.953 vs. 0.953; β_3 : food service 0.107 vs. 0.118 asymptotic, retail 0.387 vs. 0.385) – confirms rather than overturns the Model A and Model C results above, under an inference method built specifically for this few clusters.

6.2 Model Diagnostics

Model A diagnostics summary

industry	n_obs	n_influential	max_cooks_distance	shapiro_p_value
food_service	1260	78	0.0455	0
retail	1260	84	0.0265	0





Residuals are non-normal by Shapiro-Wilk (though that test is hypersensitive with 1,260 observations). More informative: the specific highest-leverage points in both industries are 2020 Q2-Q3 observations – Vermont and New York (Q2 2020), Hawaii (Q3 2020, heavily tourism-dependent) – the same COVID window Section 4 already identified, not a separate data problem. No single observation is extreme by Cook's distance.

6.3 Specification Curve

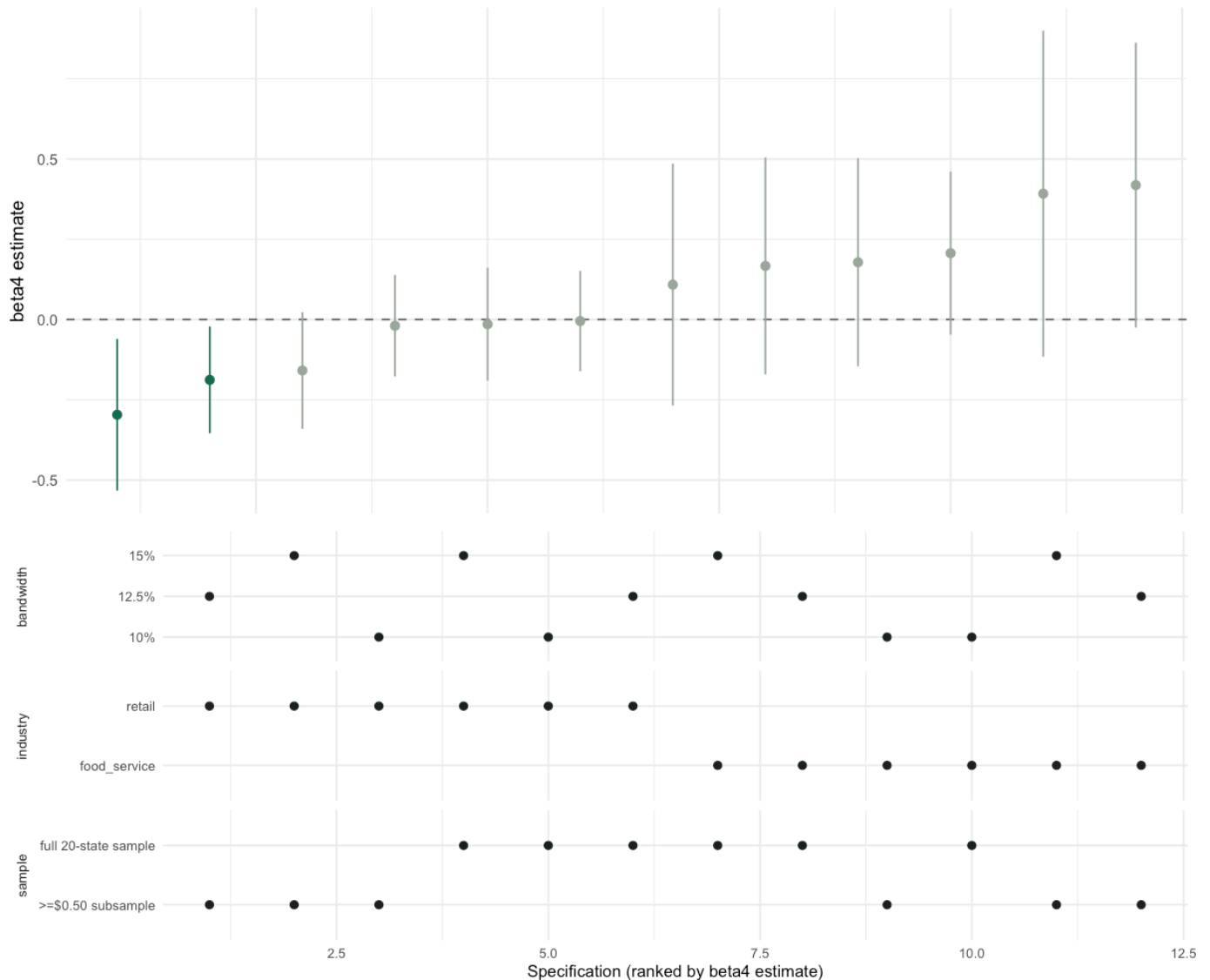
Sections 5.5 and this report's sample cut each vary one analytical choice against β_4 in isolation – exposure bandwidth alone, or the full-vs-subsample cut alone. A specification curve varies every combination of both, together with industry, in one view: 3 exposure bands x 2 samples x 2 industries = 12 specifications, all estimating the same β_4 hypothesis.

beta4 across all 12 specifications

bandwidth	sample	industry	beta4	se	p_value	significant
12.5%	>=\$0.50 subsample	food_service	0.4186	0.2264	0.0731	FALSE
15%	>=\$0.50 subsample	food_service	0.3920	0.2590	0.1394	FALSE

bandwidth	sample	industry	beta4	se	p_value	significant
10%	full 20-state sample	food_service	0.2066	0.1295	0.1177	FALSE
10%	>=\$0.50 subsample	food_service	0.1783	0.1653	0.2884	FALSE
12.5%	full 20-state sample	food_service	0.1669	0.1724	0.3384	FALSE
15%	full 20-state sample	food_service	0.1086	0.1921	0.5748	FALSE
12.5%	full 20-state sample	retail	-0.0047	0.0796	0.9530	FALSE
10%	full 20-state sample	retail	-0.0146	0.0898	0.8715	FALSE
15%	full 20-state sample	retail	-0.0194	0.0806	0.8107	FALSE
10%	>=\$0.50 subsample	retail	-0.1588	0.0926	0.0955	FALSE
15%	>=\$0.50 subsample	retail	-0.1881	0.0848	0.0334	TRUE
12.5%	>=\$0.50 subsample	retail	-0.2965	0.1205	0.0191	TRUE

Specification curve: beta4 across every exposure band x sample x industry combination
2 of 12 specifications significant at $p < 0.05$



Only 2 of the 12 specifications are significant at $p < 0.05$, and both are retail under the $\geq \$0.50$ subsample – the same subsample already flagged in Section 5.1 as the only place Model A reaches significance. Food service is never significant in any of the 6 specifications that use it, and its point estimate is positive in all 6 while retail's is negative in all 6 – the two industries don't even agree

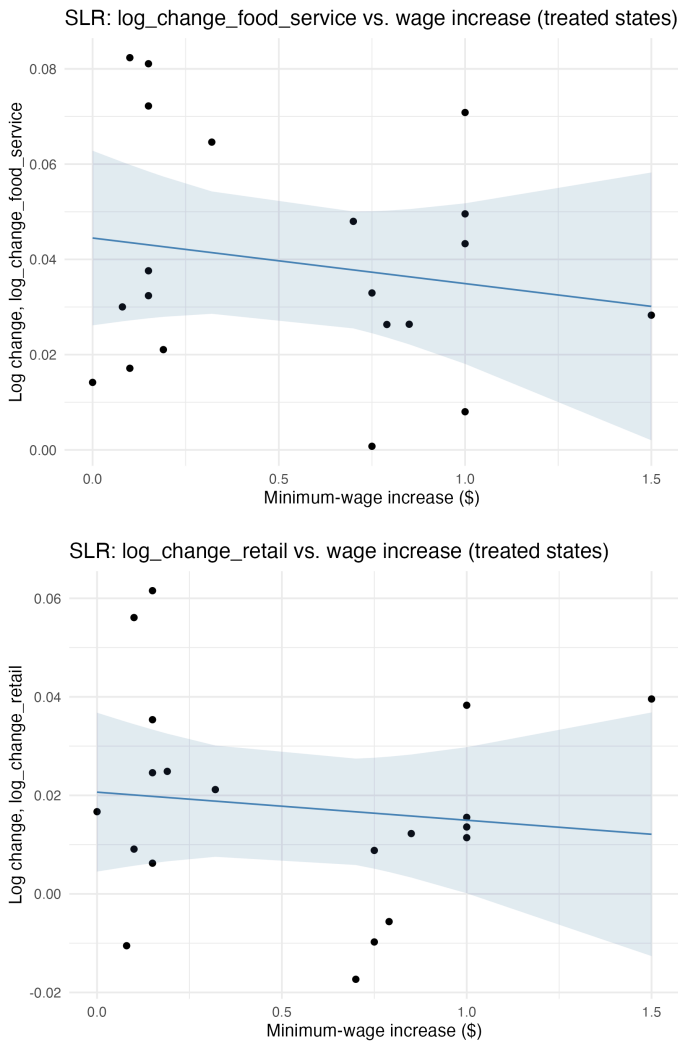
on sign. This is the clearest single piece of evidence in the report that the exposure-gradient result is fragile to analytical choices rather than a stable feature of the data.

7. Exploratory Analyses

These inform the picture but don't validate or override the confirmatory results above (Section 12 of the proposal).

7.1 SLR and MLR (STAT 240/340 coursework closure)

Simple pre/post comparisons, not the study's identification strategy. **SLR**: log employment change (2019-2020 → 2021-2022) on the dollar size of the wage increase, treated states only – not significant for either industry (food service $p = 0.46$, retail $p = 0.62$).



MLR: adds treatment status, GDP growth, population growth, and Census region, treated + control states. The treated coefficient is null for both industries (food service $p = 0.65$, retail $p = 0.26$) – an earlier version of this model that accidentally included the contaminated “excluded” group as a stand-in control showed a marginally significant result for food service; that turned out to be an artifact of the contamination, not a real signal, corrected once the real control group was built.

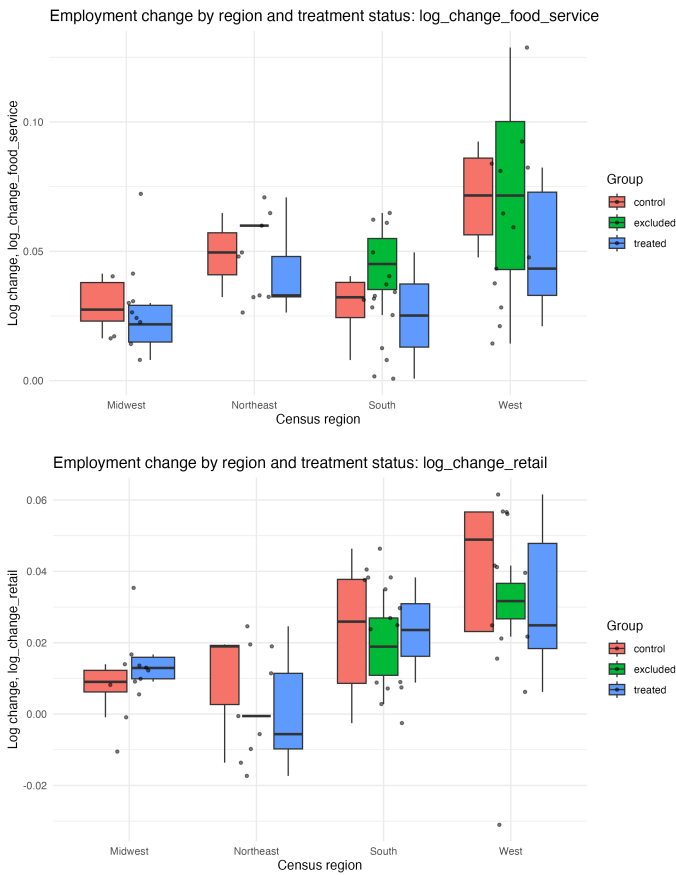
7.2 Regional ANOVA – exploratory regional variation in employment growth, not a treatment-effect estimate

This uses the full sample – treated, excluded, *and* control states together – as a general regional comparison, not a causal test of minimum-wage policy. Labeled explicitly here so it isn't mistaken for another heterogeneity check on the treatment itself: that's Model C (Section 5.2), the formal test.

One-way ANOVA: log employment change ~ Census region

outcome	f_statistic	df_region	df_residuals	p_value
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outcome	f_statistic	df_region	df_residuals	p_value
log_change_food_service	5.1044	3	46	0.0039
log_change_retail	5.6414	3	46	0.0022



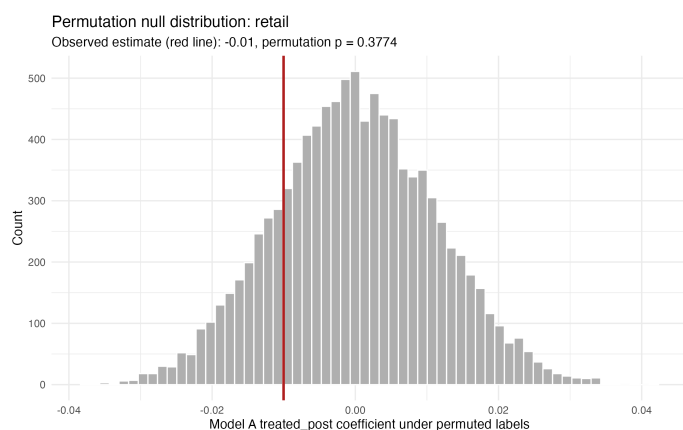
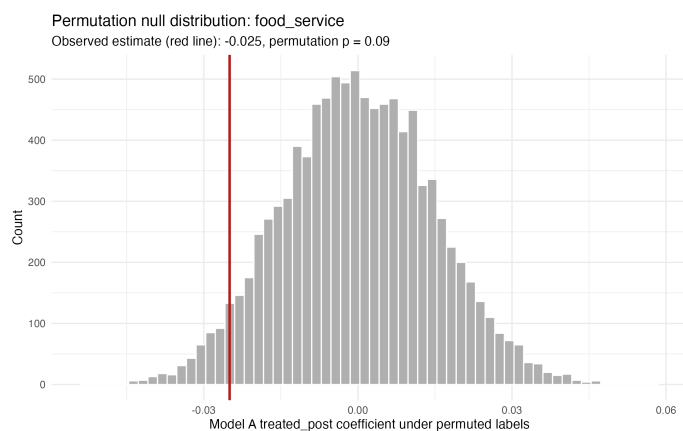
Regions differ significantly in employment growth for both industries (West notably higher than Midwest/South) – real geographic heterogeneity, unrelated to treatment. The exposure interaction in Model C remains the formal heterogeneity test; this is exploratory only.

7.3 Monte Carlo Permutation Test

10,000 relabelings of which states are “treated” (state-level, holding n_{treated} fixed at 20), Model A refit each time.

Permutation test, Model A treated_post

industry	observed_treated_post	null_mean	null_sd	permutation_p_value
food_service	-0.025	0	0.0147	0.0900
retail	-0.010	0	0.0112	0.3774

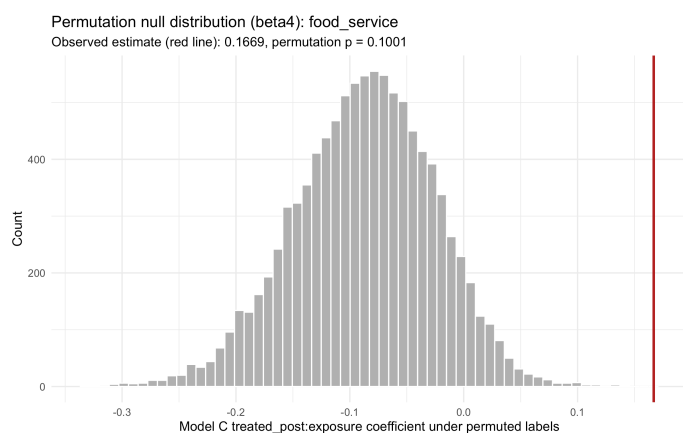


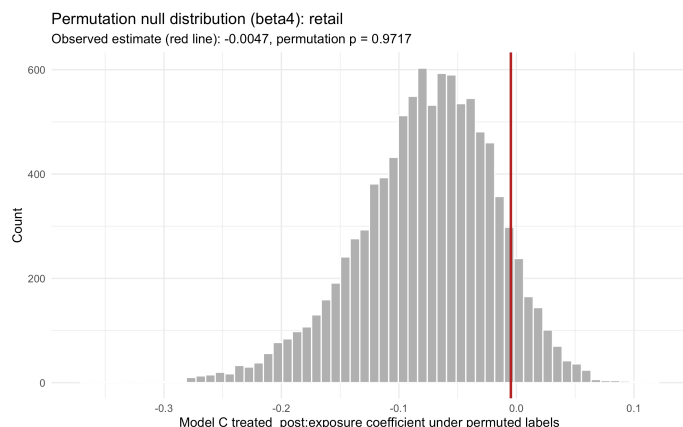
Permutation p-values track the asymptotic Model A p-values closely (food service 0.09 vs. 0.118; retail 0.377 vs. 0.385) – another confirmation of the Model A result via a completely different inferential route.

Post-build addition: this section originally covered Model A only. The same relabeling exercise applies to Model C’s exposure interaction (β_4 , Section 5.2, the primary scientific hypothesis) – each replication keeps every state’s own exposure value fixed and only randomizes which states count as “treated,” the same logic as the Model A version above.

Permutation test, Model C (beta4) treated_post:exposure

industry	observed_beta4	null_mean	null_sd	permutation_p_value
food_service	0.1669	-0.0872	0.0617	0.1001
retail	-0.0047	-0.0767	0.0586	0.9717





Retail's permutation p-value (0.972) tracks its asymptotic one (0.953) about as closely as Model A's do. Food service's don't: permutation $p = 0.100$ versus an asymptotic $p = 0.338$ – more than a 3x difference, though neither crosses the conventional 0.05 threshold. The reason is visible in the null distribution itself: its mean is -0.087, not approximately zero the way Model A's permutation null is (0.00001). Reassigning which states are “treated” while leaving each state's real, fixed exposure value in place produces a systematically negative placebo β_4 on average – exactly the kind of exposure-linked, treatment-unrelated structure Section 4.4's placebo test already found at a false date. This is now the third independent check (with the specification curve and the placebo test) pointing at the same underlying issue: part of what looks like an exposure gradient in this data may be a pre-existing association between exposure and employment trends, not a treatment effect.

7.4 Power Analysis for β_3 (Model A average effect, added after the original build)

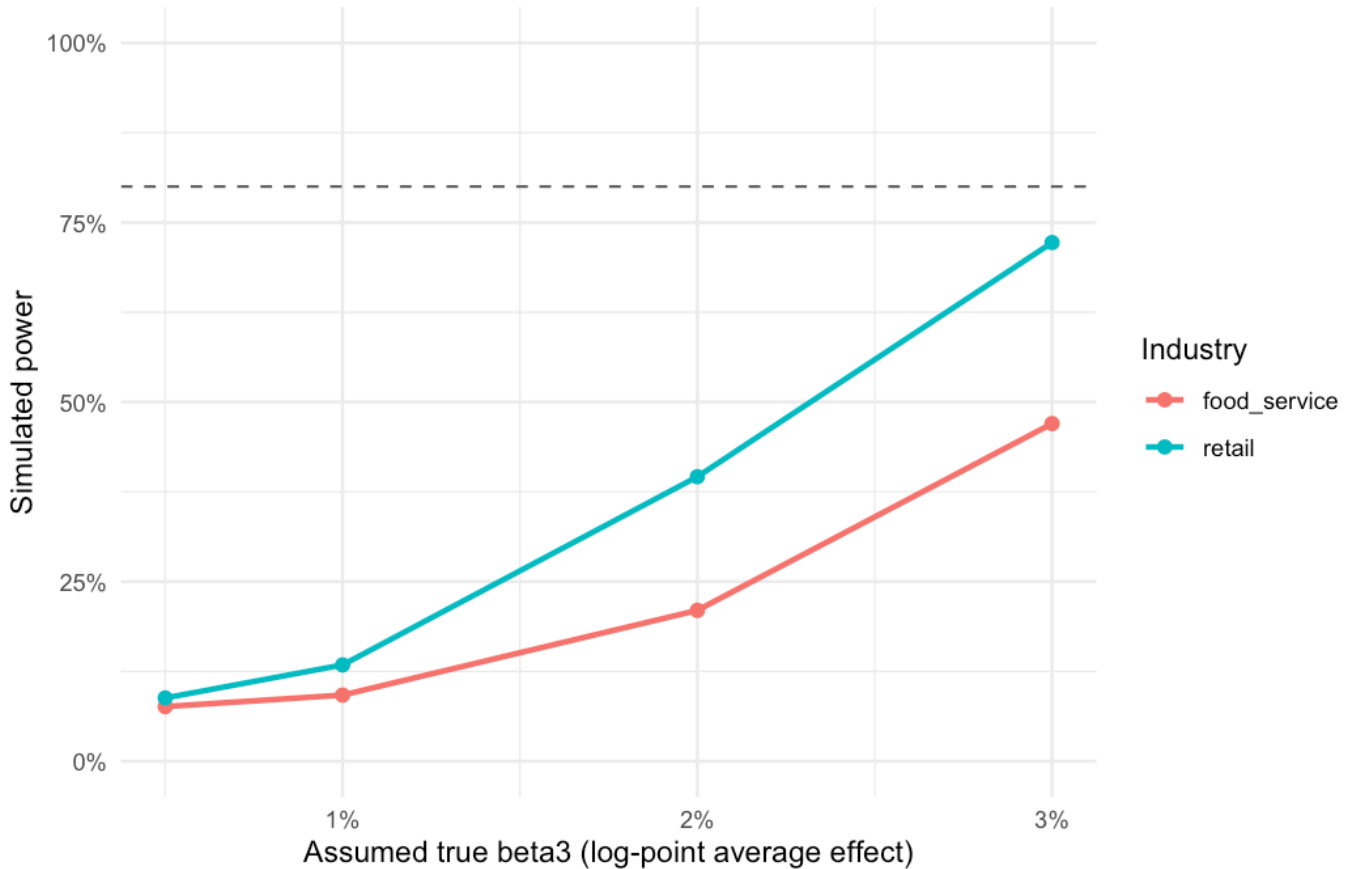
Section 4.4 specifies two power analyses, not one: average-effect power for β_3 at assumed sizes of 0.5%, 1%, 2%, and 3%, reporting the minimum detectable effect (MDE); and a separate heterogeneity-detection power analysis for β_4 (Section 7.5). Both were added after the original 10-day build (`R/16_power_analysis.R` ; see `TIMELINE.md`).

Observed beta3 vs. minimum detectable effect at 80% power, Section 4.4's assumed grid (0.5%-3%)

industry	observed_beta3	mde_80
food_service	-0.025	NA
retail	-0.010	NA

Power to detect beta3 (Model A average effect), by assumed effect size

Section 4.4's own assumed grid (0.5%-3%), 20 treated + 25 control states



The MDE isn't reached anywhere in Section 4.4's own assumed range for either industry – the design needs a true average effect larger than 3% before it reliably clears 80% power. That's consistent with the rest of this report: Model A only came back significant in the $\geq \$0.50$ -increase subsample (Section 5.1). Food service's estimate there (4.75%) is comfortably past the range this grid covers; retail's (2.74%) sits inside it and only barely cleared significance ($p = 0.049$) – exactly what falling short of 80% power in that region would predict, a real but not guaranteed chance of detecting it. The full 20-state sample's smaller, more diluted average effect never had a realistic shot at significance given this design's power.

7.5 Power Analysis for β_4 (exposure gradient, added after the original build)

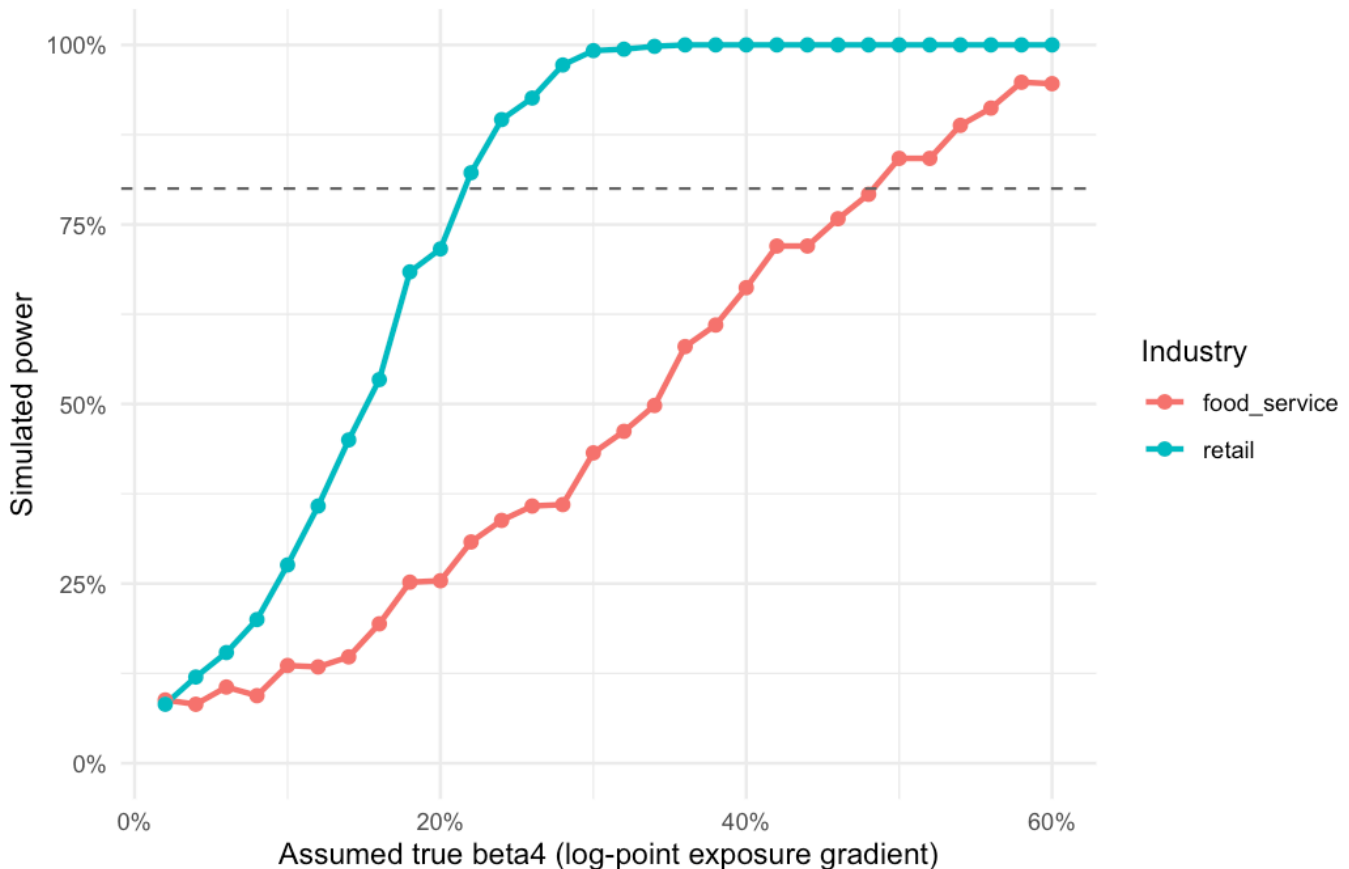
The exposure-gradient hypothesis (β_4) was separately pre-registered as likely underpowered (Section 4.4's heterogeneity-detection question). The quantitative version – a residual-based Monte Carlo power simulation on the real panel – makes that concrete.

Observed beta4 vs. minimum detectable effect at 80% power

industry	observed_beta4	mde_80
food_service	0.167	0.50
retail	-0.005	0.22

Power to detect β_4 (exposure gradient), by assumed effect size

Design: 20 treated + 25 control states, quarterly 2016-2022, state-clustered residual simulation



Both industries' observed β_4 estimates fall well short of their own minimum detectable effect at 80% power – the design would need a true exposure gradient roughly 3x (retail) to 5x (food service) larger than what was actually estimated before it could reliably distinguish that effect from noise. This is consistent with, and quantifies, the qualitative pre-registered expectation. **The simulation quantifies which effect sizes this design could realistically distinguish from zero** – it is a design property, not a way of explaining away the noisy β_4 result above after the fact.

7.6 Exploratory Treatment-Intensity Model

Exploratory, not confirmatory – not part of the six-check confirmatory family in Section 8. The dollar size of a state's increase is not randomly assigned; states that raised wages by \$1.00 vs. \$0.08 may differ systematically in ways unrelated to the policy itself. With that caveat: does a *larger* increase correspond to a larger employment response, rather than a simple treated/not-treated split?

Exploratory: log employment $\sim$ (wage increase x post), continuous dollar amount

industry	beta1_wage_increase_post	beta1_se	beta1_p
food_service	-0.0358	0.0172	0.0429
retail	-0.0196	0.0140	0.1689

Food service's coefficient is negative and marginally significant ($p = 0.04$); retail's is negative but not significant. Directionally consistent with Model A's binary result, but given the non-random assignment of increase size, this doesn't upgrade the evidence – it's one more piece of exploratory information, not a sharper confirmatory estimate.

8. Multiple Testing and Inferential Hierarchy

Two things are true at once in this report, and they're not in tension: Model A (Section 5.1, β_3) is the headline result, and β_4 (Section 5.2) is the primary scientific hypothesis. They answer different questions – “is there an average effect” versus “does it vary with exposure” – and one is never substituted for the other. Model A is reported first because it's the best-powered result (Section 7.4

quantifies exactly how well-powered); β_4 is reported alongside its own power ceiling (Section 7.5) rather than read as a clean null just because it came back noisy.

The **confirmatory family** is small and explicit, fixed before any model was run rather than assembled after seeing results: Model A and Model C for both industries (Sections 5.1, 5.2), the event study (Section 4.1), the exposure-measurement validation (Section 2), both power analyses (Sections 7.4, 7.5), the wild cluster bootstrap (Section 6.1), and the placebo/COVID-sensitivity checks (Sections 4.3, 4.4). This family is what determines the headline claim.

The **regional ANOVA** (Section 7.2), the **SLR/MLR** coursework-closure pieces (Section 7.1), and the **Monte Carlo permutation test** (Section 7.3) are exploratory. For Model A they inform the picture without changing it – the permutation test corroborates β_3 's asymptotic p-value from a distribution-free angle. For β_4 , the permutation test's role is different: its null distribution for food service isn't centered at zero, which is itself evidence about the exposure measure rather than just a robustness check on the p-value (Section 8.1 below). None of these exploratory pieces validate or overturn the confirmatory family's result, and none are treated as if they could.

8.1 Convergent Evidence Against the Exposure-Gradient Interpretation

Four independent checks, run for different reasons and living in four different sections, all point the same direction on β_4 . Collected in one place rather than left scattered:

Four independent checks on beta4's credibility

check	food_service	retail
Placebo test (false 2018 date)	$p < 0.0001$	$p = 0.0164$
Permutation test (label reassignment)	$p = 0.1001$ (null not centered at zero)	$p = 0.9717$ (null not centered at zero)
Specification curve (12 specs)	0 of 6 significant, positive in 6 of 6	2 of 6 significant, negative in 6 of 6
Event study (pre-trend joint test)	$p < 0.0001$	$p < 0.0001$

No single one of these is individually damning – a placebo test can fail by chance, a specification curve can look messy without meaning much, one permutation null can drift off-center in a finite sample. What makes this convergent rather than coincidental is that four methodologically unrelated approaches (a false-date comparison, a label-randomization null, an exhaustive grid of analytical choices, and a dynamic quarter-by-quarter decomposition) all land on the same conclusion: high- and low-exposure states differ systematically for reasons that predate the 2021 policy. β_4 is reported throughout this report exactly as designed – honestly, regardless of significance – but it should be read as **inconclusive at best, and confounded by a pre-existing exposure-outcome association at worst**, not as evidence either for or against a real exposure gradient in the employment response.

Naming the confirmatory family also means it can be corrected for formally, not just described narratively. Holm-adjusted p-values across the 4 primary tests (β_3 and β_4 , both industries, headline specification):

Holm-adjusted p-values, confirmatory family

coefficient	industry	p_raw	p_holm
beta3	food_service	0.1179	0.4717
beta3	retail	0.3853	1.0000
beta4	food_service	0.3384	1.0000
beta4	retail	0.9530	1.0000

None of the 4 survive correction at the headline specification – the same conclusion Sections 5 and 7 already reach individually, now stated as a formal joint test rather than four separate uncorrected p-values that happen to agree.

9. Limitations

- **Non-random policy adoption**; parallel trends is a tested assumption, not a proven one, and Section 4 above shows it doesn't fully hold for food service.
- **COVID confound**: the single largest limitation this analysis found. A meaningful share of the baseline effect is attributable to differential COVID recovery, not minimum wage policy.

- **Limited power**, for both β_3 and β_4 – pre-registered in Section 4.4 of the proposal as a qualitative expectation for both, and borne out by the actual results. The formal quantitative piece for both was scoped out of the original 10-day build, but was added afterward (`R/16_power_analysis.R`, a residual-based Monte Carlo simulation on the real panel). For β_4 : the minimum detectable effect at 80% power is $\beta_4 \approx 0.50$ for food service and $\beta_4 \approx 0.22$ for retail, both well above the actual observed estimates (0.167 and -0.005). For β_3 (Model A, Section 7.4): the MDE isn't reached anywhere in the proposal's own assumed 0.5%-3% range for either industry – consistent with Model A's significant results only showing up in the $\geq \$0.50$ -increase subsample, where the estimated effects are large enough to fall in or past that range. The qualitative pre-registration held up for both parameters, and the numbers that would have quantified it going in now exist too.
- **Exposure-measure construction risk**: band width (10-15%) and cell pooling rules are documented choices, not the only defensible ones.
- **The exposure-gradient placebo test fails** (Section 4.4): at a false 2018 treatment date, the exposure-interaction term is significant for both industries, meaning high- and low-exposure states already had different employment trends before 2021 for reasons unrelated to the real policy. This directly threatens β_4 's identification specifically – part of the real estimate may reflect this pre-existing structural difference rather than a treatment effect, consistent with Section 6.3's specification curve showing the same result is fragile to sample and industry choice. Section 7.3's permutation test corroborates this from a third, unrelated angle: its placebo null distribution for β_4 isn't centered at zero for food service (mean -0.087) the way Model A's permutation null is – reassigning treatment labels while keeping each state's real exposure value fixed produces a systematically negative “effect” on average, the same pattern the placebo test found.
- **Multi-change states excluded** from the primary comparison. The conflation of small legislated increases with automatic CPI adjustments turns out not to be separable at all in this data, not just “only partially addressed”: the $\geq \$0.50$ -increase subsample and the legislated-only subsample are the exact same 10 states (confirmed in `data/processed/model_a_c_results.csv`'s `n_legislated_treated` column), so the dollar-threshold cut and the mechanism cut can't be disentangled here – any difference attributed to “legislated vs. automatic” in this sample is equally attributable to “large vs. small.”
- **~20-45 clusters**: addressed via state-clustered SEs throughout and the wild cluster bootstrap for β_4 , but inference with this few clusters always carries some caution.
- **Spillovers between neighboring states** aren't modeled – a control state bordering several treated states could see labor migrate or employers adjust in anticipation, which would bias the control group's outcome toward the treatment effect (attenuating the estimate) rather than leaving it a clean counterfactual.
- **Anticipation effects**: employers and workers may have responded ahead of the 1/1/2021 effective date once increases were signed into law or scheduled, which the binary treated/post cutoff doesn't capture.
- **External validity**: this window is COVID-era by construction: even setting the COVID-sensitivity confound (above) aside, an estimate from 2019-2022 may not generalize to a minimum-wage increase in a more typical labor market.
- **The permutation test (Section 7.3) is a reference distribution, not evidence of a randomized experiment**. States chose whether and when to raise wages; permuting treatment labels tests statistical extremeness against a null of no effect, not against a null of “as-if-random” assignment, since the assignment was never random to begin with.

10. Conclusion

The working hypothesis going in was a small average effect with an inconclusive exposure gradient, reported honestly regardless of significance (Section 15 of the proposal). The average-effect half held up largely as expected: a real, COVID-driven confound the proposal's own credibility-check infrastructure was built to catch (Sections 4.1-4.3), plus a power ceiling that explains why Model A was only significant in the $\geq \$0.50$ -increase subsample (Section 7.4). Success here isn't a clean significant result – it's that the event study, two-sample t-test, placebo test, and COVID-sensitivity spec did their jobs and surfaced a real problem before it could be reported as a clean finding.

The exposure-gradient half turned out to need more scrutiny than “inconclusive” alone would suggest. Beyond the power ceiling that independently limits what β_4 could detect (Section 7.5), four methodologically unrelated checks added after the original build – a false-date placebo test, a label-randomization permutation test, a 12-specification robustness curve, and a dynamic quarter-by-quarter event study (Section 8.1) – all converge on the same finding: high- and low-exposure states differ systematically for reasons that predate the 2021 policy. That reframes β_4 from merely underpowered to actively confounded, and the report says so directly rather than defaulting back to “inconclusive” once the stronger evidence was in hand.

11. Machine Learning & Neural Network Extension

Everything above is aimed at a causal question: did the 2021 minimum-wage increases change employment. This section asks a different, narrower question instead – how well can flexible ML methods *predict* quarterly employment growth for a state they never saw during training, versus a plain linear specification? It is a **predictive-accuracy comparison, not a causal claim**, and doesn't revise anything concluded above.

Two design choices matter for interpreting the numbers below. First, evaluation holds out entire *states*, not rows: a row-level split would let the same state's quarters appear on both sides, leaking within-state structure the causal models' clustered SEs already treat as non-independent. Second, the prediction target is employment *growth*, not the level of `log_employment`: the level is dominated by each state's sheer size (California vs. Wyoming), which none of these features could recover for a state held out entirely – that would make the held-out evaluation unwinnable by construction. Growth is scale-invariant across states and is the quantity a policy could plausibly move.

Held-out predictive accuracy, by state (R/27_ml_prediction.R, R/28_neural_network.R)

model	rmse	r_squared
linear_baseline	0.0699	0.3939
bagged_trees	0.0667	0.4476
random_forest	0.0643	0.4871
gradient_boosting	0.0536	0.6439
neural_network	0.0582	0.5796

Five methods, in increasing order of how much structure they can capture:

- **Linear baseline** – no state or quarter fixed effects (unlike Model A/C above), since a fixed effect can't generate a prediction for a state it never saw.
- **Bagged trees** – bootstrap-aggregated `rpart` trees, each with one fixed random feature subset for its entire fit (the “random subspace” method).
- **Random forest** (`randomForest` package) – the actual Breiman algorithm: every split, in every tree, re-samples which features are even candidates, not just once per tree.
- **Gradient boosting** (`gbm` package) – a different combination strategy entirely: trees are fit sequentially, each one to the residual error the ensemble so far still has, with the number of trees chosen by 5-fold cross-validation rather than fit until they stop improving on the training data.
- **Neural network** (`nnet`) – a single hidden layer; with ~1,000 training rows and 5 predictors, a deeper architecture would be capacity in search of data it doesn't have.

Gradient boosting is the strongest predictor of the five, which is a sensible result rather than a foregone one: boosting's sequential, residual-fitting strategy is genuinely different from bagging and random forests' independent-trees-then-average strategy, and there's no a priori guarantee either wins on a given dataset.

An independent Python replication (`python/`) – `numpy/pandas` for data handling, hand-rolled implementations of the same three tree ensembles plus a PyTorch feedforward network, since this development environment's `scipy/scikit-learn` install doesn't import cleanly (see `python/README.md`) – reaches the same qualitative conclusion: gradient boosting outperforms the other tree ensembles and the linear baseline. Full details, and why `BaggedTrees` / `RandomForest` / `GradientBoostedTrees` are three genuinely different implementations rather than the same ensemble under different names, are in `python/README.md` and `python/src/tree.py`.

12. Wage Pass-Through

A new research question, not a re-analysis of the one above: did the 2021 increases actually raise **earnings** in low-wage industries, and by how much relative to the dollar size of the mandate? Sections 1-11 all ask whether employment fell; this asks whether the policy did the thing it was actually meant to do.

Data limitation, found by direct request (`R/29_wage_passthrough.R` , matching this project's standing practice of confirming FRED series IDs against live data rather than guessing): BLS/FRED do not publish state-level average hourly earnings at the detailed NAICS-722 (food service) or 44-45 (retail trade) level Sections 1-11 use for employment – every candidate series ID at that resolution 404s. The finest available state-level breakdown is the CES **supersector**: “Leisure and Hospitality” (nests food service, but also accommodation and arts/entertainment/recreation) as the closest proxy for food service, and “Trade, Transportation, and Utilities” (nests retail, but also wholesale trade, transportation, and utilities) as the closest proxy for retail. Both are broader than this study's employment industries, so this section answers a related but not identical question – not “did retail workers' wages rise” but “did wages in retail's broader supersector rise.”

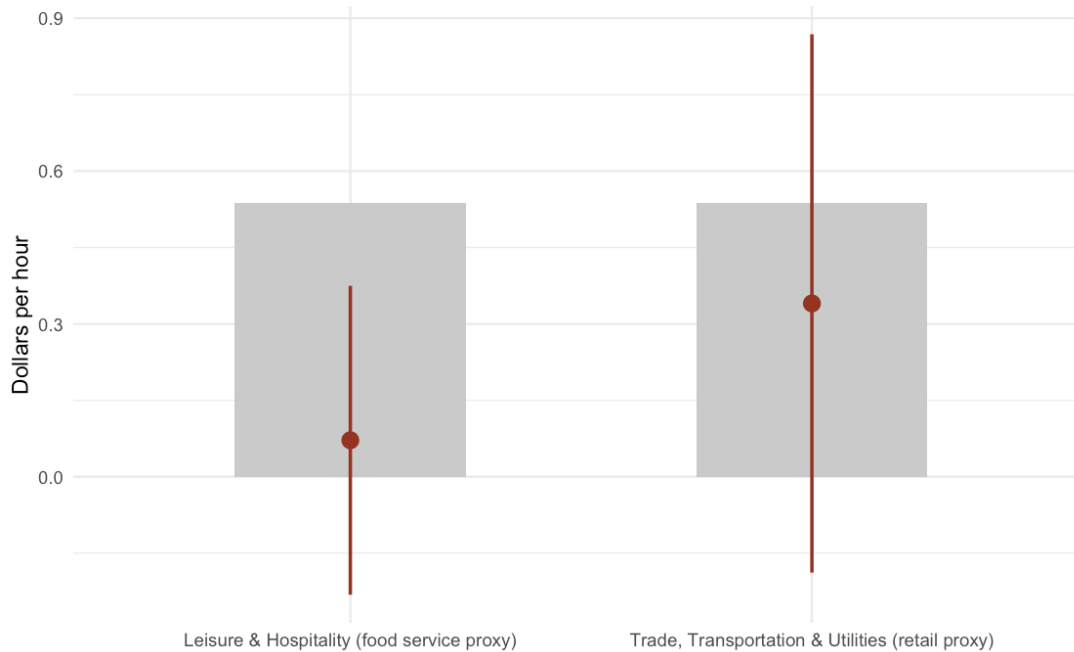
The specification is otherwise identical to Model A (Section 5.1): `build_panel()` and `fit_model_a()` from `R/07` are reused completely unchanged, with the wage-proxy series substituted in as the outcome.

Wage pass-through: Model A's treated_post on proxy-industry earnings

proxy_industry	treated_post_log_coef	se	p_value	mandated_dollar_increase	implied_dollar_increase	pass_through_ratio
Leisure & Hospitality (food service proxy)	0.0044	0.0095	0.6467	0.5365	0.0716	0.1334
Trade, Transportation & Utilities (retail proxy)	0.0144	0.0115	0.2169	0.5365	0.3402	0.6342

Wage pass-through: mandated vs. estimated dollar increase

Grey bar = avg. mandated statutory increase; red point = estimated proxy-industry wage effect (95% CI)



Neither estimate is statistically significant, and the food-service proxy's confidence interval crosses zero. Read alongside this report's recurring theme (Sections 7.4-7.5's power analysis, the noisy β_4 throughout), the honest interpretation is "underpowered to detect pass-through at this resolution," not "pass-through is genuinely 13%/63%." The likeliest reason both point estimates sit well under 100% regardless: the supersector proxies include large numbers of workers (hotel managers, arts/entertainment/recreation staff, wholesale and utilities employees) who were never near the minimum wage to begin with, diluting any real effect on the low-wage workers the policy actually targeted.

Reproducibility: every number and figure in this report is generated by running `R/01_treatment_classification.R` through `R/29_wage_passthrough.R` in order, from `min-wage-difference-in-differences-study`'s repo root (make all does this end to end). See `README.md` and `TIMELINE.md` for the day-by-day build log, including every defect found and corrected along the way.